Draft Project Report

Employee Attrition Classification

Group 73

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Problem Setting

This project highlights the significant impact of our employment experiences on our happiness, as most of our waking hours as adults are spent at work. While finding a profession one loves and is passionate about is crucial for a happy life, it's not enough to keep an employee motivated and engaged in the long run. In today's world, employers have a responsibility to ensure that their employees are satisfied, motivated, and willing to stay with the company.

To achieve this, it is essential for companies to understand the factors that influence an employee's decision to stay or leave a particular organization. The dataset being analyzed aims to quantify such factors, which can range from employee satisfaction levels to salary and even the safety of the working environment.

Employee satisfaction is a crucial factor that can impact an employee's decision to stay or leave a job. Satisfied employees tend to be more motivated, productive, and engaged, while dissatisfied employees are more likely to leave. Other factors like work-life balance, workload, opportunities for growth and development, and recognition and rewards also play a crucial role in employee satisfaction.

Another essential factor is the amount of time spent at work. Long working hours, frequent overtime, and lack of flexibility in work schedules can lead to burnout, stress, and reduced motivation among employees. Additionally, factors like salary, benefits, and job security can also influence an employee's decision to stay or leave a company.

Finally, a safe working environment is also crucial for employee retention. Employees who feel safe and secure at work are more likely to be satisfied, motivated, and productive. On the other hand, a lack of safety measures or incidents of workplace harassment or discrimination can lead to a negative work environment and drive employees away.

Overall, understanding and addressing these factors can help companies retain their employees and create a positive and productive work environment. By doing so, companies can not only improve employee satisfaction and motivation but also reduce turnover rates and improve overall performance.

Problem Definition

The objective of analyzing the dataset is to determine which variables have the most significant impact on an employee's decision to stay or leave a job. To achieve this, the data must first be cleaned to remove any errors, inconsistencies, or missing values that may affect the accuracy of the analysis.

Once the data is cleaned, machine learning models can be used to analyze the relationships between the various variables and employee retention. The goal is to identify which parameter values, such as salary, job satisfaction, and working hours, have the greatest impact on an employee's decision to stay or leave their job.

Different machine learning algorithms can be used to model and predict the outcomes based on the data. These models can be trained and evaluated based on specific metrics, such as accuracy, precision, and recall, to determine their effectiveness.

Once the models have been trained and evaluated, the best-performing model can be selected for further analysis. The selected model can then be used to identify which variables have the most significant impact on employee retention and can be used to make predictions about future employee retention rates.

Overall, by analyzing the dataset and using machine learning models, companies can gain insights into the factors that influence employee retention and make data-driven decisions to improve employee satisfaction, motivation, and retention. This can ultimately lead to a more productive and positive work environment, reduced turnover rates, and improved business performance.

Data Sources

Kaggle-

https://www.kaggle.com/datasets/giripujar/hranalytics?select=HR_comma_sep.csv

Data Description

Dataset contains 14999 records available with 10 columns.

satisfactoryLevel : Scores given by employees, ranging 0 to 1

lastEvaluation : Last evaluation points given, ranging 0 to 1

numberOfProjects : Number of projects involved

avgMonthlyHours : Average monthly hours

timeSpent.company : Time spent at the company, in years

workAccident : Whether he/she had a work accident

Left : if the employee is about to leave or not, about to leave(serving notice period)- 1 and 0

promotioninLast5years : Whether he/she had a promotion in the last 5 years

dept : Department he/she belongs to

Salary : Salary as high, medium or low

Data Preprocessing

The data consists of categorical as well as continuous variables. The categorical variables were converted from object to integer by using label encoder. Integers were assigned to the class names. The continuous variables were standardized using Standard Scaler.

```
satisfaction_level       float64
last_evaluation          float64
number_project             int64
average_montly_hours       int64
time_spend_company         int64
Work_accident              int64
left                       int64
promotion_last_5years      int64
Department                 int64
salary                     int64
dtype: object
```

Following are the statistics for the continuous variables of the data

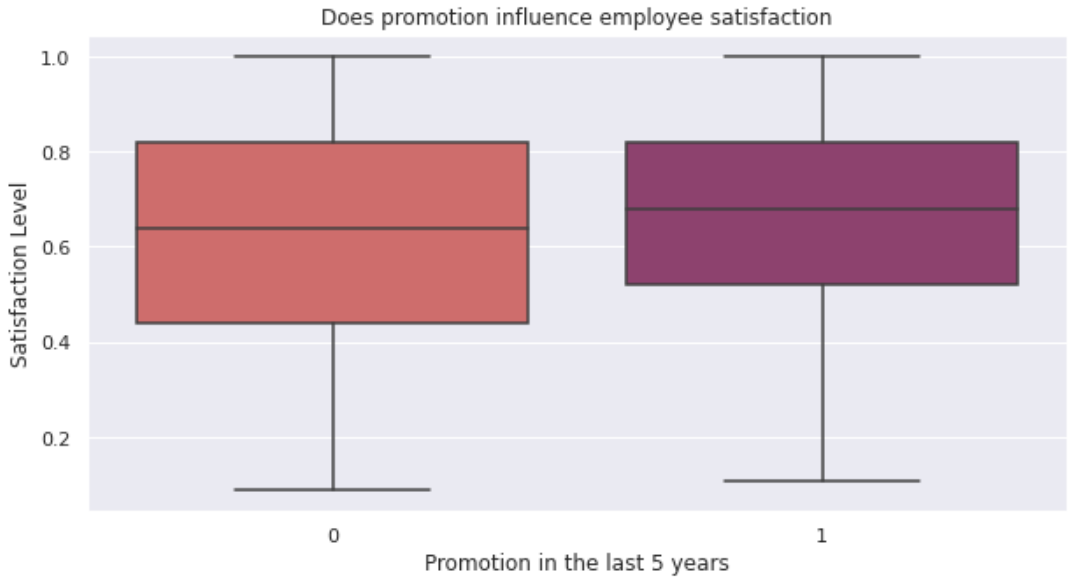
	satisfaction_level	last_evaluation	number_project	average_montly_hours	time_spend_company
count	14999.000000	14999.000000	14999.000000	14999.000000	14999.000000
mean	0.612834	0.716102	3.803054	201.050337	3.498233
std	0.248631	0.171169	1.232592	49.943099	1.460136
min	0.090000	0.360000	2.000000	96.000000	2.000000
25%	0.440000	0.560000	3.000000	156.000000	3.000000
50%	0.640000	0.720000	4.000000	200.000000	3.000000
75%	0.820000	0.870000	5.000000	245.000000	4.000000
max	1.000000	1.000000	7.000000	310.000000	10.000000

It is observed that the average ‘satisfaction_level’ is 0.61, the average number of projects they worked on is 3.80, and the average time an employee has spent in the company is 3.49 years.

The continuous variables were reduced from 5 to 4 using Principal Component Analysis, the variance statistics are given below

```
Proportion Variance [0.36592932 0.22182628 0.17542438 0.12700308]
Cummulative Proportion of Variance [0.36592932 0.58775559 0.76317997 0.89018305]
```

Data Visualization

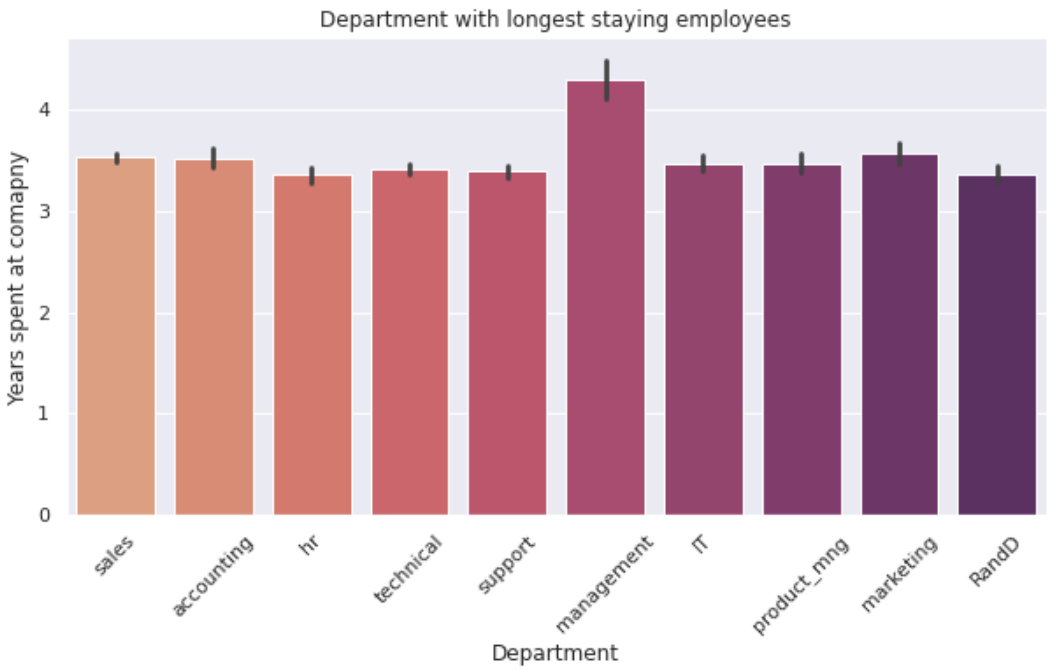


The above plot shows the impact of promotion on satisfaction level.

1: Promoted in the past 5 years

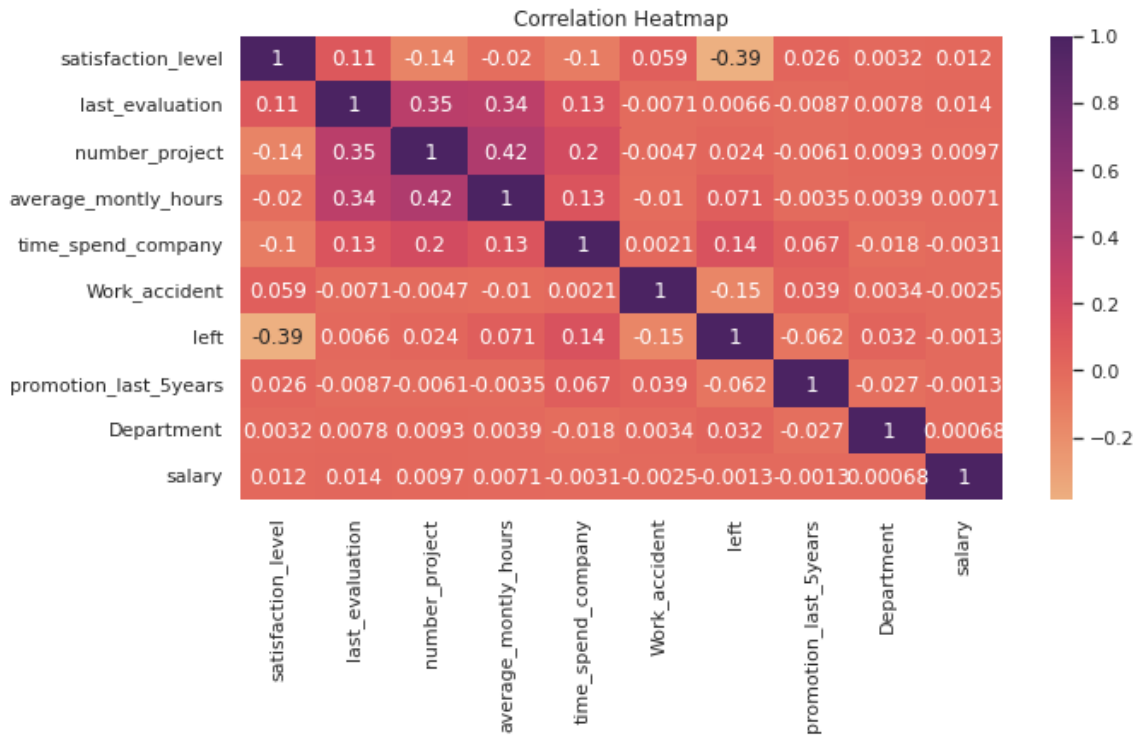
0: No promotion in the past 5 years

When we plot the relation between promotion in the last 5 years and the employee's satisfacti on, we discover that the employees who have received a promotion in the last 5 years have a higher median satisfaction level than the employees who have not received a promotion. The y also have a higher satisfaction rating in general.



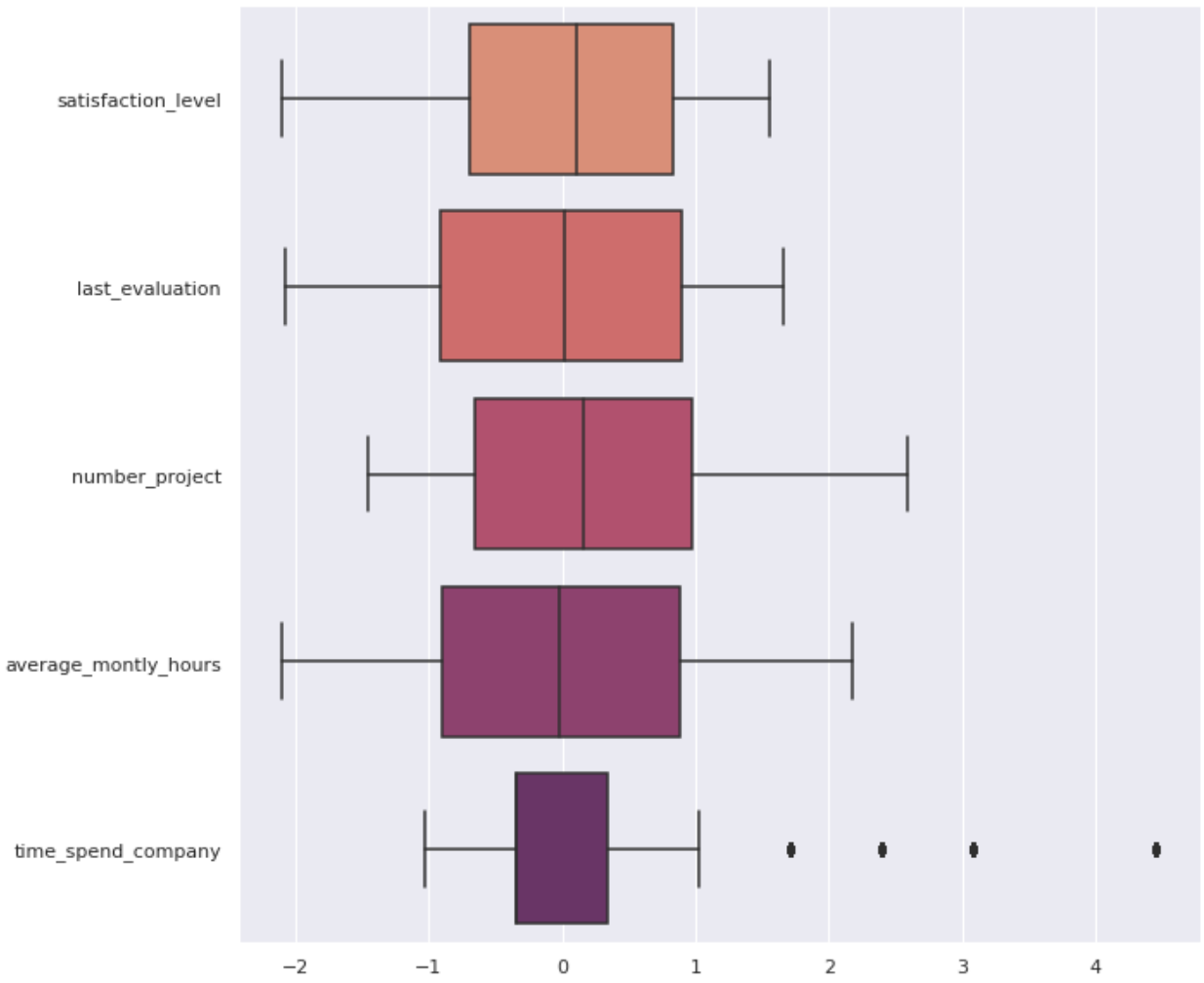
The above bar-plot aims at finding the department with the longest staying employees and as it is evident from the plot, Management has the highest retention of employees of all the departments in the company.

The above bar-plot shows the relation between employee retention and their salary. As expected, employees with a low salary have left the most whereas employees with a high salary were retained the most.



Key takeaways from the above correlation heatmap:

- Average monthly hours and number of projects the employee worked on has the highest positive correlation.
- Followed by satisfaction level of employee and the 'Left' variable. The variables are negatively correlated.
- Number of projects and last evaluation also have a significant positive correlation.
- Work accidents also seem to have an impact on employees leaving since they have a negative correlation.



The above boxplot shows the distribution of records and outliers in the data after standardization of all the continuous variables.

Data Mining Models/Methods

The data consists of 10 attributes. The attribute 'left' is the response variable with the success class being '1' which denotes that the employee left the company.

Since the response variable is a categorical variable, we will be exploring and using classification models.

The input space consists of both categorical and continuous predictor variables.

Neural Networks

- The data split for training the model will be:

  Train: 70% of data

  Validation: 20% of data

  Test: 10% of data

  The data is split in a way to keep enough data for validating the model while training it.
- The optimizer used will be 'adam'. Since the response variable has only 2 classes the loss function used for training will be 'binary_crossentropy'.
- The evaluation metric used while compiling the model will be 'accuracy'.
- Pros: neural networks are quick to output the predictions and can work with both categorical and numerical variables.
- Cons: training these models can be resource intensive and if the hidden layers are not set properly, they can often lead to overfitting.

Random Forest

- Random forest consists of several decision trees (the number of trees can be set before compiling the model). Each tree outputs a prediction and the most frequent occurring prediction is the one output by the model.
- Data split ratio for model training:

  Train: 75% of data

  Test: 25% of data

  Random Forest works well with large training sets as well as small training sets. Hence the smaller train split for this model.
- Pros: It is robust to outliers in the data and produces high accuracy

- Cons: It can be resource intensive compared to other learning algorithms since it takes longer to train and can be difficult to interpret.

Support Vector Machine

- Data split ratio for model training:
  Train: 80% of data
  Test: 20% of data
- The categorical predictor variables will be label encoded using the scikitlearn python library.
- Pros: It is less prone to overfitting and works well with small sample sizes
- Cons: However, it is not scalable can performs poorly on large datasets

Logistic Regression

- Logistic Regression is usually used for binary classification problems. In the case of employee attrition rate prediction, logistic regression can be used to predict the probability that an employee will leave the company, based on various factors such as job satisfaction, salary, tenure, etc.
- Logistic Regression can handle categorical variables, which are common in employee attrition.
- Data split ratio for model training:
  Train: 80% of data
  Test: 20% of data
- Pros: It is not resource intensive and can be trained quickly, it is also quite easy to interpret
- Cons: However, it can have a high bias if the predictor and response variables have a non-linear relationship. Also, it performs poorly on multi-class classification tasks.

Model Performance Evaluation and Interpretation

Neural Networks

The data was split into 3 parts. 70-20-10 for training, validation and testing respectively. The reason for this split is that we wanted enough data for training and validation. However, to get an accurate measure of the model's performance, we kept 10 percent of test data that was never shown to the model.

```
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 dense_40 (Dense)            (None, 64)                640

 dense_41 (Dense)            (None, 32)                2080

 dense_42 (Dense)            (None, 16)                528

 dense_43 (Dense)            (None, 1)                 17

=================================================================
Total params: 3,265
Trainable params: 3,265
Non-trainable params: 0
```

Model Architecture

The activation function used for the first 3 layers was 'relu' and the output layer had 'sigmoid' as the activation function with 1 node, since this is a classification task.
The loss function was 'binary_crossentropy' for the same reason.

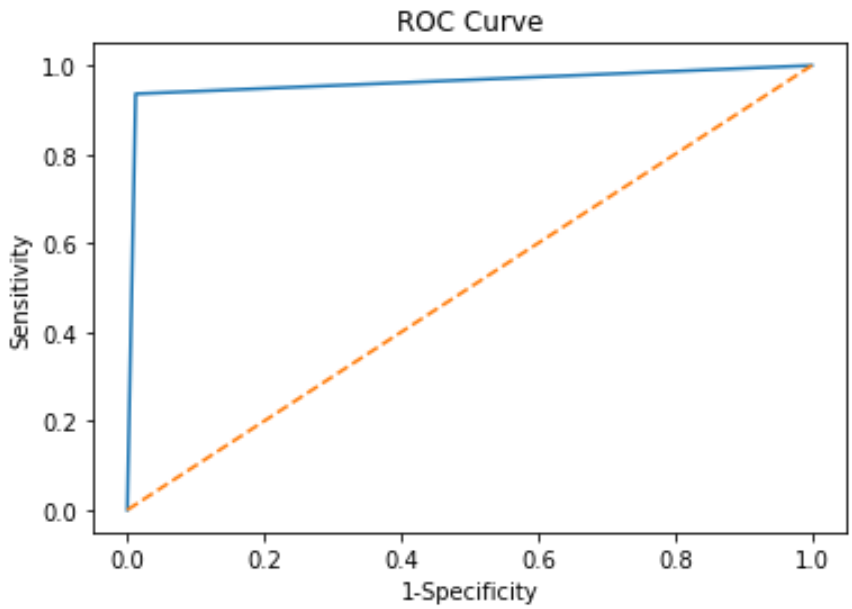
With epochs = 10

Data	Accuracy
Train	97.16%
Validation	96.41%
Test	96.53%

Epochs = 20

Data	Accuracy
Train	96.81%
Validation	96.48%
Test	96.93%

	Predicted C1	Predicted C2
Actual C1	307	38
Actual C2	8	1147

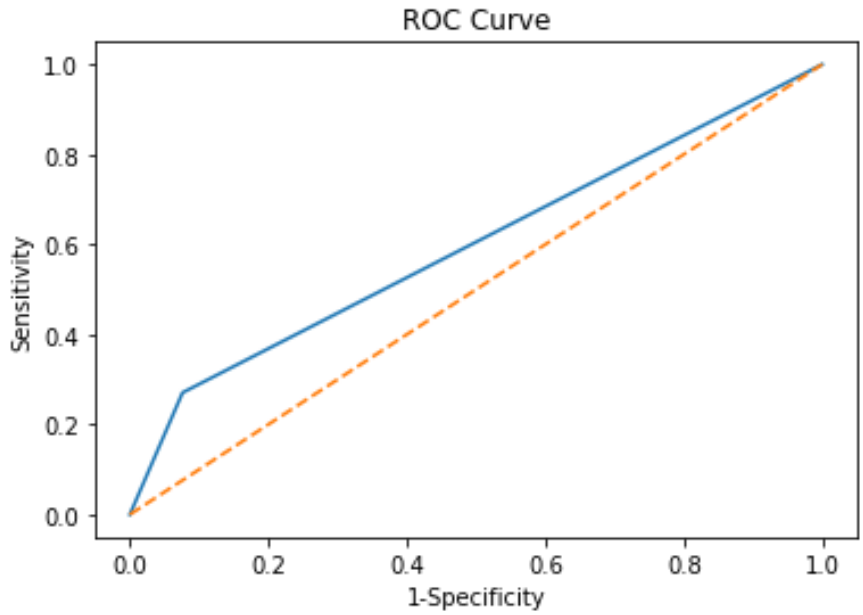


Logistic Regression

The data was split into 80% training and 20% testing as a standard split ratio

Accuracy on the test data was 76%, which is quite poor compared to the neural networks model.

	Predicted C1	Predicted C2
Actual C1	190	523
Actual C2	179	2108

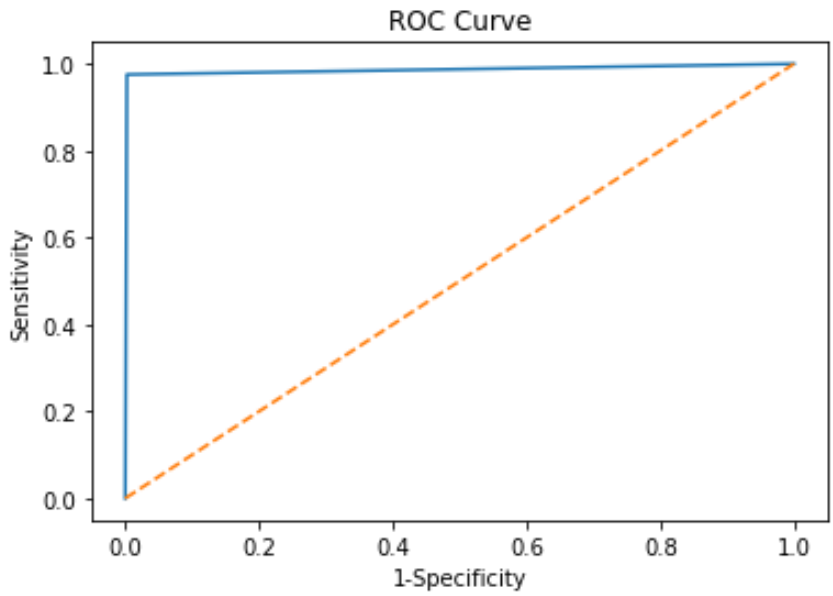


Random Forest

The data was split into 75% training and 25% testing. The reason for smaller training split is that decision trees take relatively longer to run.

The random forest model had an accuracy of 99.22% on the test dataset.

	Predicted C1	Predicted C2
Actual C1	866	31
Actual C2	18	2835

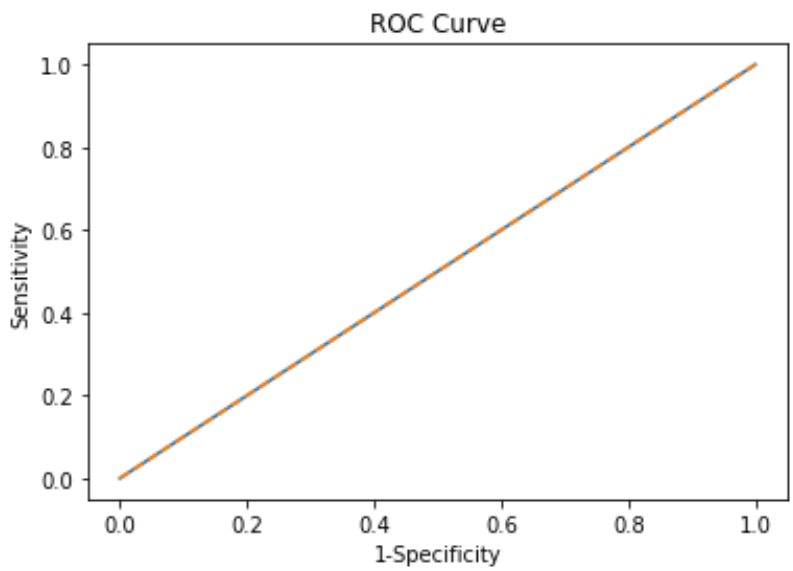


Support Vector Machine

The data was split into 80% training and 20% testing as a standard split ratio.

Accuracy on the test data was 76%

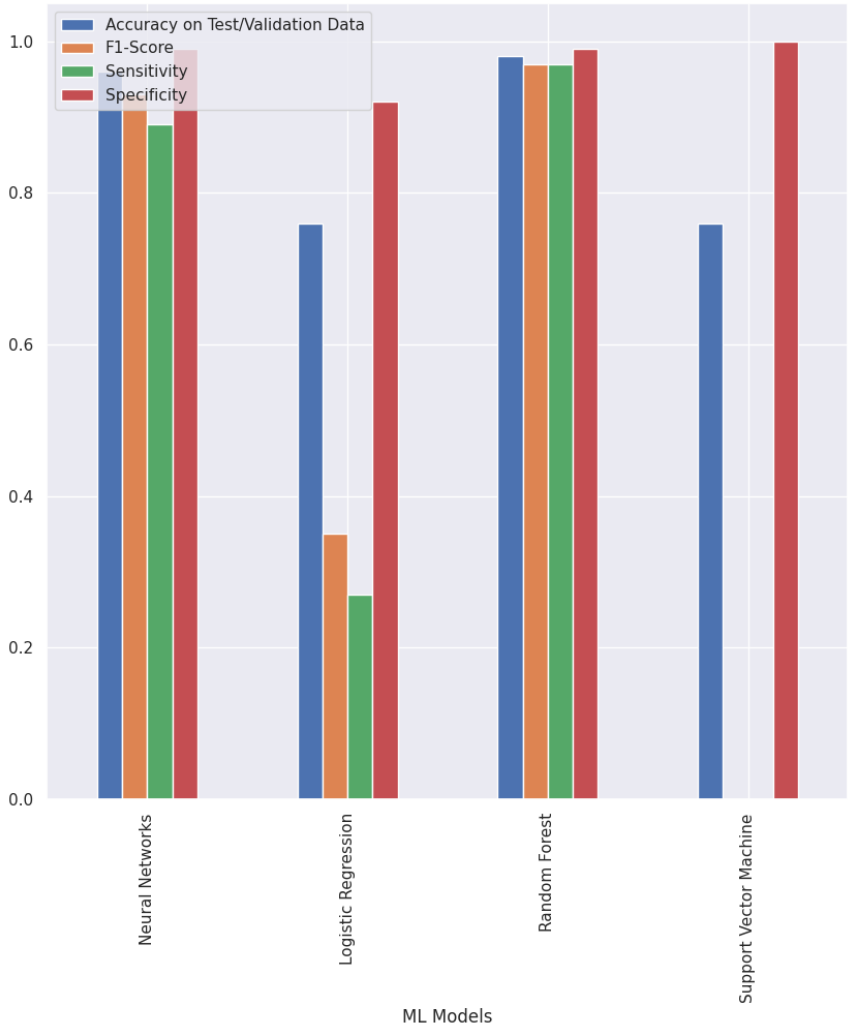
	Predicted C1	Predicted C2
Actual C1	0	713
Actual C2	0	2287



Performance Comparison

Model	**Accuracy on Test/Validation Data**	**F1-Score**	**Sensitivity**	**Specificity**
Neural Networks	0.96	0.93	0.89	0.99
Logistic Regression	0.76	0.35	0.27	0.92
Random Forest	0.98	0.97	0.97	0.99
Support Vector Machine	0.76	0.0	0	1

Based on the above results and ROC curves, Random Forest is the best model due to the high accuracy and F1 score. SVM is the worst classifier, since it has an F1 Score of 0 and the ROC Curve is the textbook version of a poor classifier.



Project Results

The objective of this machine learning project was to identify the most significant variables that impact an employee's decision to stay or leave a job. The first step involved cleaning the dataset to remove any inconsistencies, missing values, or errors that could affect the accuracy of the analysis.

Once the data was cleaned, we used various machine learning algorithms such as neural networks, logistic regression, support vector machine (SVM), and random forest to model and predict the outcomes based on the data. The goal was to identify which parameter values, such as salary, job satisfaction, and working hours, had the greatest impact on an employee's decision to stay or leave their job.

To evaluate the effectiveness of these algorithms, we used specific metrics such as accuracy, F1-Score, sensitivity and specificity. The models were then trained and evaluated on the dataset to determine their effectiveness in predicting employee retention rates.

After training and evaluation, we found that the random forest algorithm performed the best, with the highest accuracy in predicting employee retention rates.
This is largely due to it’s robustness to outliers and good performance on large datasets such as ours. The poor performance of Logistic was due to it’s assumption of linearity between dependent and independent variables and the poor performance of SVM was due to the size of the dataset, since SVM performs poorly on large datasets.

Impact of Project Outcomes

The impact of this machine learning project can be significant for both employees and employers. For employees, identifying the factors that contribute to job satisfaction and retention can lead to a more fulfilling work life. Employees can expect better working conditions, improved communication with management, and potentially higher salaries if employers use data-driven decisions to retain their staff.

For employers, the project's impact can be substantial in improving retention rates and reducing costs associated with employee turnover. When employers take proactive measures to create a positive work environment, they can retain top-performing employees, reduce recruitment costs, and increase productivity levels. The project can also help employers identify potential problems in the work environment and take corrective measures before it leads to employee attrition.

The machine learning models developed in this project can provide predictive insights for employers, helping them anticipate and prevent employee turnover. Employers can use this information to invest in employee development programs, create a more inclusive work environment, and provide other benefits and incentives to improve retention.

Overall, the project's impact can improve the working lives of employees and lead to a more productive, positive, and cost-effective work environment for employers.